Linear Regression

1. Univariate Linear Regression (LR)

2. Multivariate Linear regression

3. Find the parameters of LR

   → Cost function

   → Gradient Descent.

Machine Learning Flow (Any algorithm)

A: Find Data → B: Load Data → C: Data preprocessing → D: Split data → E: Train model on data → F: Evaluate model

1.1 Load data

1.2 Explore data

C1

1.3 Selecting variable

- Train data
- Testing data

1.3 Prepare data

- Algorithm
- 1.4 Train

maths here

- 1.5 Evaluate

C2: Exploratory Data Analysis

C3: Feature Engineering → missing values, Encoding

One HOT ENCODING

C4: Feature Scaling

How to look at a dataset

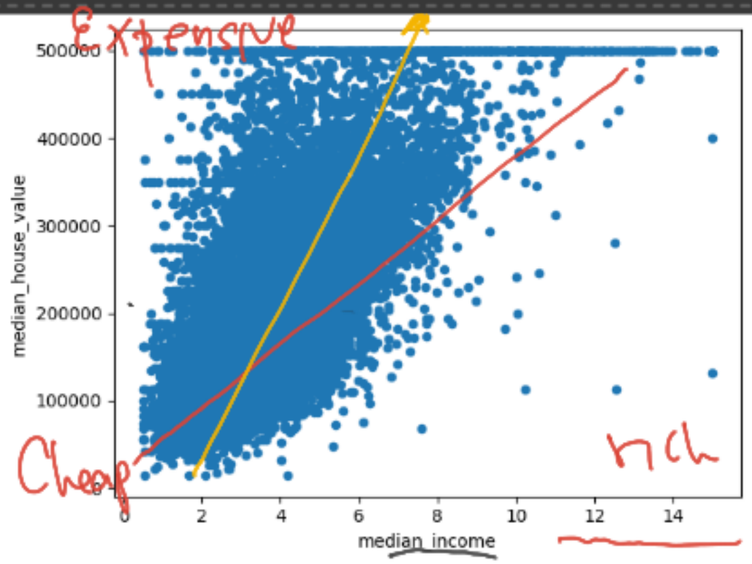
Features / independant variables, attributes

```
df = pd.read_csv('california_housing_train.csv')
df.head()
```

target output / target var

	x_1 longitude	x_2 latitude	x_3 housing_median_age	x_4 total_rooms	x_5 total_bedrooms	x_6 population	x_7 households	x_8 median_income	median_house_value
0	-114.31	34.19	15.0	5612.0	1283.0	1015.0	472.0	1.4936	66900.0
1	-114.47	34.40	19.0	7650.0	1901.0	1129.0	463.0	1.8200	80100.0
2	-114.56	33.69	17.0	720.0	174.0	333.0	117.0	1.6509	85700.0
3	-114.57	33.64	14.0	1501.0	337.0	515.0	226.0	3.1917	73400.0
4	-114.57	33.57	20.0	1454.0	326.0	624.0	262.0	1.9250	65500.0

rows = samples / observation



Hypothesis:

H_1: More rooms = expensive house

H_2: Old houses = expensive "

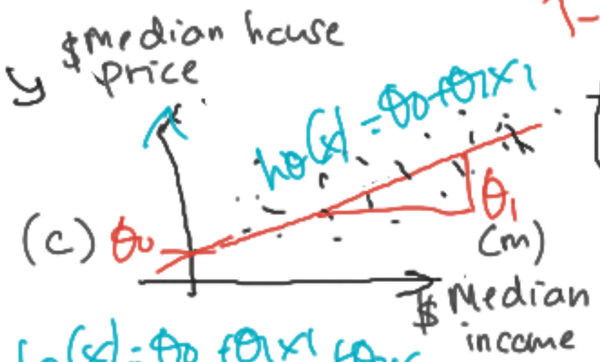
H_3: Higher income = expesive house

Linear Regression (LR)

↳ 1) (Univariate) One variable LR

2) (Multivariate) Multi variable LR

In High dimensions (>3D) $h_\theta(x)$ is called hyperplane.



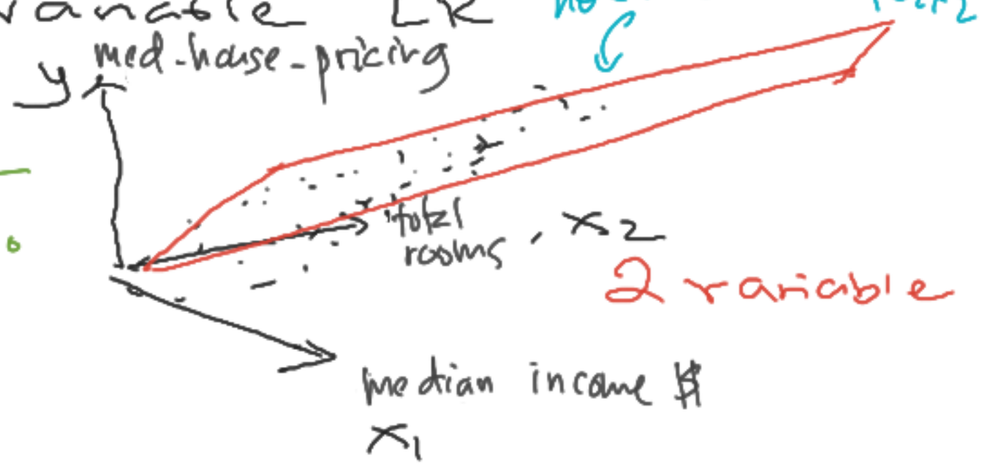
1- variable

$= c + mx$

$$\boxed{y = mx + c}$$

$h_\theta(x) = \theta_0 + \theta_1 x_1$

intercept, slope



2 variable

$$\boxed{y = m_1 x_1 + m_2 x_2 + c}$$

$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$

Q How about 8 variables? What does the eqn looks like?

A) $h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_8 x_8$

(θ_1: m_1, θ_2: m_2, θ_3: m_3, θ_8: m_8)

Linear Regression (LR)

- We are given this california house.csv dataset.
- We want to do some LR on this dataset.
- For example. Can we predict

  What is the predicted price of a house given that we know the house has a total of 7 rooms.

Revision of Linear Algebra

x , $x = 5$: scalar

$$\mathbf{x} , \mathbf{x} = \underbrace{\overbrace{\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}}^{3\ \text{col}}}_{1\ \text{row}} \rightarrow \mathbf{x} \text{ is a } \overset{\text{row} \times \text{col}}{1 \times 3} \text{ row vector}$$

$$\mathbf{x} = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{1\ \text{col}} \text{ (column) vector}$$

3 rows

$\mathbf{x}$ is a $\overset{\text{row} \times \text{col}}{3 \times 1}$ column vector

$\mathbf{X}$ is a 3×3 matrix

$$\mathbf{X} = \underbrace{\begin{bmatrix} 1 & 10 & 100 \\ 2 & 20 & 200 \\ 3 & 30 & 300 \end{bmatrix}}_{3\ \text{row}} \Big\} 3\ \text{col}$$

Continue

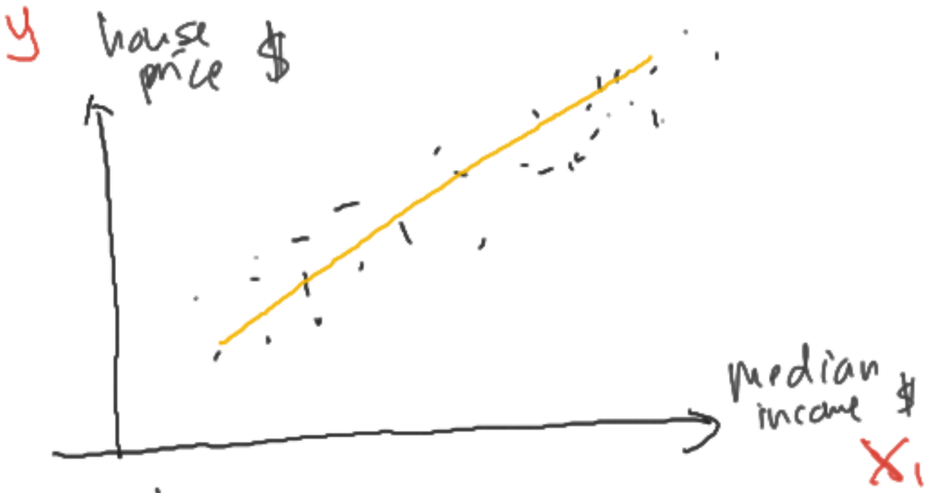
$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix} \quad \mathbf{x}_3 = \begin{bmatrix} 100 \\ 200 \\ 300 \end{bmatrix}$$

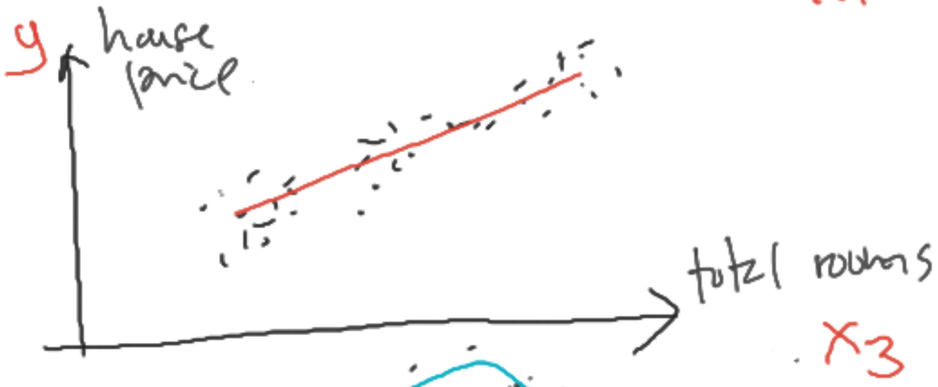
$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 10 & 100 \\ 2 & 20 & 200 \\ 3 & 30 & 300 \end{bmatrix}$$

Our 'intuition'

x_1 x_2 x_3 x_4 x_5 x_6 y





x: features/attributes, independent variable

y: target output, dependant variable

y Charges

Linear Regression with one variable

	x median income	y house price
1	1.4	\$66.9k
2	1.8	\$.80k
⋮		
?	1.6	?

- Linear regression is to find a linear line that fits the dataset.
- We can make predictions for unseen observation. For example what is the house price for a person with income = 1.6?

y
Charges

How do we find the best line (red) that fits the data?

$$y = \underline{m}x + \underline{c}$$

$$h(x) = \theta_0 + \theta_1 x$$

c: intercept m: slope

- So our job (for one variable) is to solve for $\underline{m}$ and $\underline{c}$.

θ_1 θ_0

How do we find the 'best'
θ_1 θ_0
m and c ?

$$y = mx + c \qquad h_\theta(x) = \theta_0 + \theta_1 x_1$$

Two methods

A) Solve using equation → the Ordinary Least Square method (OLS)
→ the normal equation (High dimensional...)

B) Use gradient descent

A) Solve using equation

Ordinary Least Square method (from Stats)

θ_0 c, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x};$

$y = mx + c$

$\hat{\beta}_0$: Estimate of intercept (c) θ_0

$\hat{\beta}_1$: Estimate of the slope (m) θ_1

m, θ_1 $\hat{\beta}_1 = \dfrac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2},$

$y = \beta_0 + \beta_1 x$

$h_\theta(x) = \theta_0 + \theta_1 x_1$

what is this?

From our dataset

index	x_3 median income	y house price
1	1.4	\$66.9k
2	1.8	\$80k
i	$x_i =$	$y_i =$

$n =$

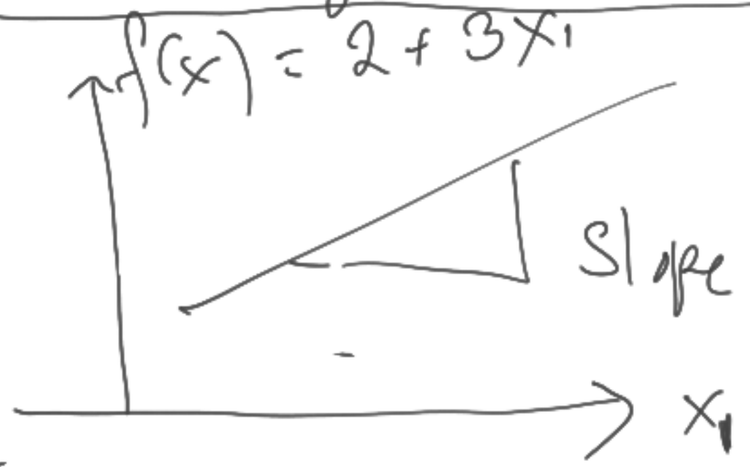
$\bar{x}$: averge of x $= \frac{1}{N}\sum_{i=1}^{N} x_i$

$\bar{y}$ = " of y $= \frac{1}{N}\sum_{i=1}^{N} y_i$

Revision (Calculus)

What is a gradient?

rate of change



Univariate

$$\text{Grad } f(x) = \frac{df(x)}{dx} = 3$$

$$f(x) = 2 + 3x_1^2 + 10x_2^2$$

Multivariate

A vector of gradients

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \end{bmatrix}$$

partial derivatives

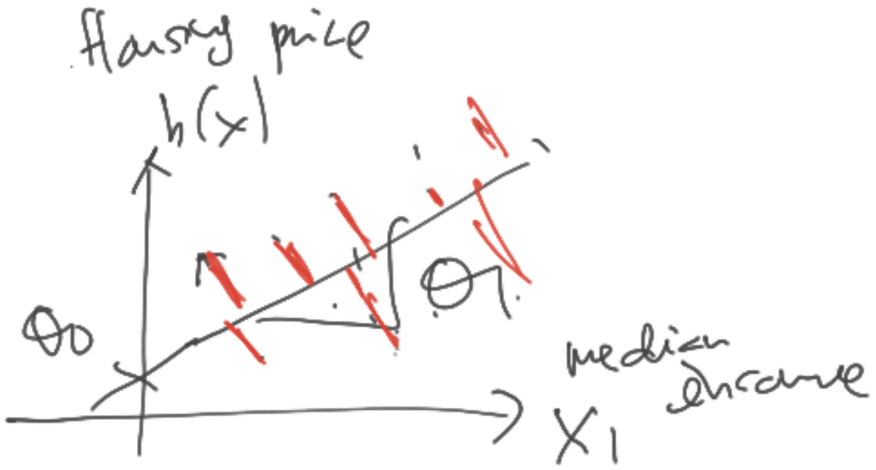
$$\nabla f(x) = \begin{bmatrix} 6x_1 \\ 20x_2 \end{bmatrix}$$

3.) Use gradient Descent

(gradient of the cost function)

We need to define a cost function

$h(x) = \theta_0 + \theta_1 x_1$ ← our red line



● : our data ● : our prediction (line)

$$\left(h(x^i) - y^i\right)^2 = \left(\underbrace{[\theta_0 + \theta_1 x_1^i]}_{h(x)^i} - y^i\right)^2$$

$\theta_0 = 2$
$\theta_1 = 3$

∴ So our cost function is

$$J(\theta_1, \theta_0) = \frac{1}{2N} \sum_{i=1}^{N} \left(h(x^i) - y^i\right)^2$$

indx	X income	y house price	h(x)	$(h(x) - y)^2$
1	1.4	\$66.9k		
2	1.8	\$80k		
⋮				
i	1.6	\$75k		
⋮				

x^i y^i $h(x^i)$ 32,000

N =

$$\underset{\theta_0, \theta_1}{\text{minimize}} \; \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

m: #training examples

$h_\theta(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

$$\underset{\theta_0, \theta_1}{\text{minimize}} \; \underbrace{J(\theta_0, \theta_1)}_{\text{Cost function}}$$

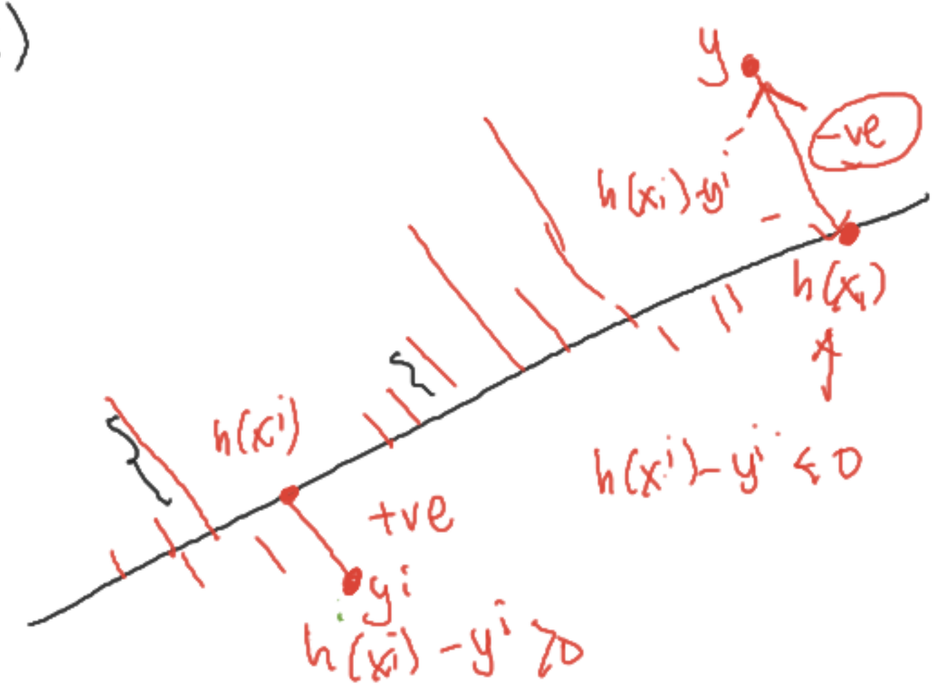
Squared error function

3.) Use gradient Descent

(gradient of the cost function)

Cost function $\quad h(x) = mx + c$

$$J(m,c) = \frac{1}{2N} \sum_{i=1}^{N=138} \left(h(x^i) - y^i \right)^2$$



- The cost function calculates the distance between our predictions $h(x)$ and the original data.

- So then we need to find the values of m and c that minimizes the cost function J